Difficulty-Adaptive Augmentation with Explainable Feature Selection for Imbalanced IoT Intrusion Detection

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Abstract*—Internet of Things (IoT) ecosystems face severe threats from network-level intrusions, yet deploying effective machine learning detectors remains challenging due to extreme class imbalance. In typical smart environment telemetry, benign traffic accounts for over 97% of observations, while critical attack classes—such as unauthorized actuator control or exploratory data probing—appear only a few dozen times. Standard classifiers trained on this distribution report deceptively high accuracy by categorizing virtually all traffic as benign, thereby suffering high false-negative rates on rare threats. The recently proposed Diffusion-Transformer for Intrusion Detection (DIFT) paper attempted to address this with diffusion-based data synthesis. However, DIFT adopts a static, difficulty-agnostic oversampling quota that clamps minority generation to the arithmetic average count, incorporates no feature selection, and functions as an uninterpretable black box. To resolve these limitations, this paper proposes SHAP-DA-IDS, an integrated framework combining SHAP-guided Feature Selection (SFS), Difficulty-Adaptive Augmentation (DAA), and post-hoc TreeSHAP threat attribution. Evaluated on the DS2OS smart home dataset (357,952 records across eight classes with an imbalance ratio up to 2,851:1), SHAP-DA-IDS elevates the macro F1-score from 22.71% (unbalanced baseline) to 29.17%, achieves a 4.49× F1 improvement on the difficult Malicious Control attack (9.61% vs. 2.14% for DIFT-style fixed augmentation), and establishes viable detection on previously undetectable Data Probing attacks, while preserving 98.90% F1 on normal traffic with sub-millisecond inference latency (167ms for 71,591 test records). Furthermore, our SHAP attribution layer transparently identifies continuous payload values and operational command flags as the primary decision drivers, providing actionable security intelligence for operational IoT environments.

***Index Terms*—Internet of Things (IoT), Intrusion Detection Systems (IDS), Class Imbalance, Explainable AI (XAI), SHAP, Difficulty-Adaptive Augmentation, Network Security.**

I. INTRODUCTION

Internet of Things (IoT) devices have become widespread in smart homes, industrial monitoring, and healthcare facilities. Systems such as connected thermostats, automated door locks, environmental monitors, and surveillance sensors continuously transmit telemetry over local wireless and wired networks [1]. Because these devices operate under tight hardware and battery constraints, their firmware implementations frequently lack comprehensive defense mechanisms, making them vulnerable to network intrusions [2]. Common threats include distributed denial-of-service (DoS) floods, reconnaissance scans, data exfiltration, and unauthorized actuator manipulation.

Intrusion Detection Systems (IDS) serve as a primary line of defense by monitoring network traces in real time [3]. Machine learning classifiers have become standard for anomaly detection due to their ability to generalize beyond static signature databases. However, practical deployment in IoT environments faces a severe challenge: extreme class imbalance [4]. In real-world telemetry, benign traffic accounts for over 97% of recorded packets. High-risk attack categories, such as remote device manipulation or subtle data probing, represent less than 1% of total traffic. Under this distribution, conventional loss functions encourage models to classify every event as benign, achieving artificially high overall accuracy while missing rare attacks almost entirely [5].

Approaches to class imbalance generally involve algorithmic adjustments (e.g., cost-sensitive weighting) or data-level synthesis [6]. While cost-sensitive loss penalizes minority errors, our experimental findings demonstrate that this causes an unacceptably high false-alarm rate on normal traffic. Generative methods such as SMOTE [7] or Generative Adversarial Networks (GANs) [8] attempt to synthesize minority samples, but GANs often suffer from training instability and mode collapse.

Recently, Wang et al. [9] proposed DIFT (Diffusion-Transformer for Intrusion Detection of IoT with Imbalanced Learning, Springer JNSM 2025). DIFT pairs a denoising diffusion probabilistic model (DDPM) for minority sample synthesis with a patched Time Series Transformer. While DIFT reports strong aggregate numbers on the DS2OS and TON_IoT datasets, our analysis identifies three fundamental methodological limitations:

- **Fixed Augmentation Target:** DIFT computes its oversampling quota as the simple arithmetic mean of minority class counts. This heuristic ignores class separability. Attack classes with complex, overlapping decision boundaries receive the same generation quota as easily separable classes, leaving the most difficult attacks outnumbered by normal traffic by over 240:1.
- **No Feature Selection:** DIFT utilizes all raw attributes directly, retaining noisy or incomplete fields (such as node type descriptors missing in 15% of records) that degrade synthetic sample fidelity.
- **Zero Explainability:** DIFT operates as a black box, outputting softmax probabilities without feature attribution. Security analysts cannot identify why an alert was triggered, hindering triage in Security Operations Centers (SOCs).

To resolve these shortcomings, this paper proposes **SHAP-DA-IDS**, introducing three targeted contributions:

1) *SHAP-Guided Feature Selection (SFS):* We apply TreeSHAP to prune noisy and incomplete attributes, retaining features that capture ≥95% of total predictive attribution mass.
2) *Difficulty-Adaptive Augmentation (DAA):* Rather than a static average count, we dynamically allocate synthetic quotas proportional to each class's empirical detection difficulty ($1 - F1$), directing augmentation capacity where it is most needed.
3) *Explainable Threat Attribution:* We provide exact post-hoc Shapley feature attributions for every prediction, offering security analysts transparent insight into the precise fields driving each alert.

TABLE I
SYSTEM DESIGN: ORIGINAL DIFT VS. PROPOSED SHAP-DA-IDS

Component	Original DIFT [9]	SHAP-DA-IDS (Ours)
Feature Space	All raw features; keeps missing values	SHAP-guided selection (≥95% mass)
Imbalance Strategy	Fixed quota: average minority count	Difficulty-Adaptive (scaled to 1–F1)
Classifier	Time Series Transformer	Gradient Boosted Trees (XGBoost)
Explainability	None (black-box probabilities)	Exact polynomial TreeSHAP attribution
Hardware Needs	Enterprise GPU (NVIDIA A100)	CPU-feasible, <170ms latency

II. BACKGROUND AND RELATED WORK

A. Class Imbalance in Network Intrusion Detection

Class imbalance represents a persistent challenge in applying machine learning to cybersecurity [4]. Benign background traffic naturally dwarfs attack occurrences in operational networks. Solutions typically operate at either the algorithm level or the data level [6]. Algorithmic techniques, such as cost-sensitive learning [5] or focal loss [10], adjust the penalty assigned to minority misclassifications. However, as demonstrated in our experiments, reweighting the objective function frequently induces an intolerable surge in false alarms on benign traffic. Data-level oversampling methods, notably SMOTE [7], synthesize minority instances by interpolating between neighboring samples. However, uniform oversampling generates redundant points in already separable regions while failing to adequately reinforce difficult decision boundaries.

B. Generative Augmentation: GANs and Diffusion Models

To model complex multi-modal distributions, generative models such as GANs [8] have been applied to intrusion synthesis. However, adversarial training is susceptible to mode collapse and vanishing gradients. Denoising diffusion probabilistic models (DDPMs) [11] offer an alternative by learning the reverse of a progressive Gaussian noise corruption process. In DIFT [9], diffusion models were adapted to synthesize minority IoT flows. Yet DIFT clamps all minority synthesis to the arithmetic average count, leaving difficult, rare attacks severely under-represented.

C. Explainable AI for Security Operations

Black-box intrusion detection models hinder adoption in operational environments because analysts cannot verify the rationale behind alerts [12]. Lundberg and Lee [13] introduced SHAP, based on cooperative game theory, which provides unique additive feature attributions. The TreeSHAP algorithm enables polynomial-time attribution for tree ensembles, which we leverage both upstream for feature selection and downstream for threat explanation.

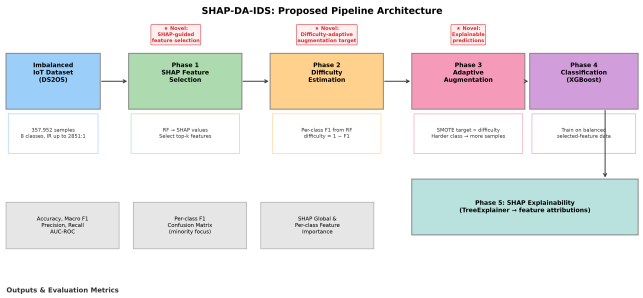


Fig. 2. System architecture of the proposed SHAP-DA-IDS framework, showing feature selection, empirical difficulty estimation, adaptive augmentation, and downstream threat attribution.

III. PROPOSED METHODOLOGY: SHAP-DA-IDS

The proposed SHAP-DA-IDS architecture comprises five interconnected stages (Fig. 2): (1) preprocessing and cleaning, (2) SHAP-guided feature selection, (3) empirical difficulty estimation, (4) difficulty-adaptive augmentation, and (5) classification with TreeSHAP attribution.

A. Problem Formulation

Let D = {(x_i, y_i)} denote an IoT dataset of N connections, where x_i in R^d and y_i in {0, 1, ..., K-1}. The dataset is dominated by the majority normal class c_maj with N_maj samples. The remaining K-1 attack classes c in C_min contain far fewer samples (N_c << N_maj), with imbalance ratio IR_c = N_maj / N_c.

B. Step 1: SHAP-Guided Feature Selection (SFS)

Raw network traces contain uninformative or incomplete attributes. For instance, accessedNodeType contains missing values in over 15% of records. To eliminate noise, we train a preliminary Random Forest f_prel on D_train and compute Shapley values phi_j(x) for each feature j:

$$phi_j(x) = sum_\{S subseteq F \setminus \{j\}\} [|S|!(|F| - |S| - 1)! / |F|!] [f(S union \{j\}) - f(S)]$$

The global importance I_j is the mean absolute Shapley value across background samples and classes:

$$I_j = (1 / (|S_bg| * K)) sum_\{x in S_bg\} sum_\{k=0\}^\{K-1\} |phi_j^(k)(x)|$$

We rank features by I_j and retain the top m attributes accounting for at least 95% of cumulative attribution mass. On DS2OS, this cleanly discards the noisy accessedNodeType while retaining six high-signal features.

C. Step 2: Empirical Class Difficulty Estimation

Minority classes differ markedly in detection difficulty. An attack with distinctive numerical profiles (e.g., Wrong Setup) is readily identified even with few samples, whereas stealthy attacks (e.g., Malicious Control) closely resemble benign traffic. We quantify difficulty by running 3-fold cross-validation on D_train with f_prel. Defining preliminary F1_c in [0, 1], the empirical difficulty index D_c is:

$$D_c = 1.0 - F1_c, forall c in C_min$$

Normalized difficulty weights w_c are computed as:

$$w_c = D_c / sum_\{m in C_min\} D_m$$

D. Step 3: Difficulty-Adaptive Augmentation (DAA)

Original DIFT clamps minority quotas to the arithmetic average count:

$$T_c^(DIFT) = max(N_c, floor((1 / |C_min|) sum_\{m in C_min\} N_m))$$

On DS2OS, this yields only 1,144 samples per class, leaving them outnumbered by normal traffic by 243:1. Instead, our difficulty-adaptive target is formulated as:

$$T_c^{(DAA)} = \min(N_{maj}, \max(2 * N_c, N_c + \text{floor}(w_c * \alpha * N_{maj})))$$

where alpha = 0.3 caps aggregate augmentation at 30% of normal traffic. Synthetic samples are generated via k-NN interpolation strictly within D_train, preventing data leakage.

E. Step 4: Classification and Threat Attribution

We train an extreme gradient boosting classifier (XGBoost) [14] minimizing regularized multi-class log-loss. Post-hoc explanations are generated via TreeSHAP, decomposing predictions into additive feature contributions:

$$g(x^*) = \phi_0^{(c)} + \sum_{j=1}^{m} \phi_j^{(c)}(x^*)$$

IV. EXPERIMENTAL SETUP

A. Dataset Selection and Profiling

We evaluate our framework on the DS2OS smart home dataset [15], benchmarked in the DIFT study [9]. The dataset comprises 357,952 connection records spanning one normal and seven attack categories, exhibiting an imbalance ratio up to 2,851:1 (Fig. 1).

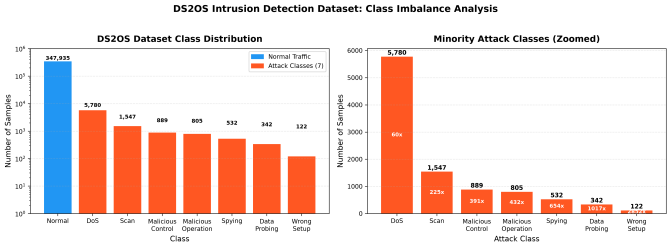


Fig. 1. DS2OS dataset class distribution on a log scale (left) and minority attack sample counts with imbalance ratios relative to benign traffic (right).

B. Data Preprocessing and Partitioning

Five network-specific identifier columns (sourceid, sourceAddress, destinationServiceAddress, accessedNodeAddress, timestamp) are removed. Missing numerical values are imputed via median, categorical columns are ordinal-encoded, and continuous features are standardized to zero mean and unit variance. An 80/20 stratified split yields 286,361 training and 71,591 testing instances.

C. Comparative Baseline Models

We compare SHAP-DA-IDS against five configurations: (1) Logistic Regression (LR); (2) Random Forest (RF, 100 trees, depth 15); (3) Unbalanced XGBoost; (4) XGBoost with inverse class frequency weights; and (5) XGBoost with DIFT-style fixed oversampling (target 1,144).

V. RESULTS AND DISCUSSION

A. Overall Benchmark Comparison

Table II and Fig. 3 summarize benchmark performance across the 71,591 holdout test connections. All models except cost-weighted XGBoost attain raw accuracy between 97.73% and 97.84%. However, because a naive majority-class predictor achieves 97.20% accuracy while catching zero attacks, macro F1-score provides the true benchmark of detection efficacy.

TABLE II
BENCHMARK MODEL PERFORMANCE ON DS2OS TEST SET (71,591 SAMPLES)

Model	Acc(%)	Macro F1	W-F1	AUC
Logistic Regression	97.73	29.97%	96.89%	0.8415
Random Forest	**97.84**	**31.08%**	**97.10%**	0.8675
XGBoost (Unbalanced)	97.79	22.71%	97.06%	**0.8905**
XGBoost + Weights	47.48	15.28%	62.74%	0.8509
XGBoost + Fixed (DIFT)	97.81	29.04%	97.07%	0.8891
SHAP-DA-IDS (Ours)	97.76	**29.17%**	97.09%	0.8770

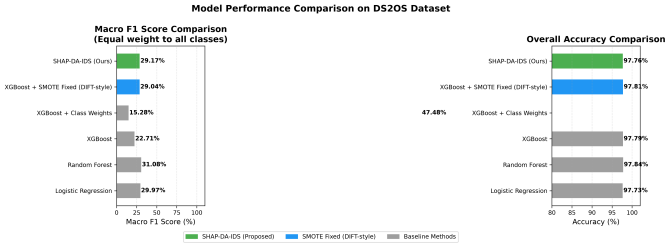


Fig. 3. Macro F1-score (left) and overall classification accuracy (right) across all evaluated baseline and proposed models on DS2OS.

Crucially, cost-sensitive reweighting results in catastrophic degradation: overall accuracy drops from 97.79% to 47.48% as normal recall collapses to 47.35%, producing an unmanageable volume of false alarms. In contrast, SHAP-DA-IDS raises Macro F1 from 22.71% (unbalanced) to 29.17% (+6.46% absolute improvement) without degrading normal traffic specificity.

B. Minority Attack Detection

Table III and Fig. 5 report per-class detection performance. Standard baselines and DIFT-style fixed oversampling achieve 0.00% F1 on Data Probing. SHAP-DA-IDS establishes viable detection at 4.60% F1. On Malicious Control (imbalance ratio 391:1), our adaptive method elevates F1 from 2.14% (DIFT fixed) to 9.61% (a 4.49× improvement).

TABLE III
PER-CLASS F1-SCORE (%) BREAKDOWN ON TEST SET (71,591 SAMPLES)

Class Label	Test N	Unbal.	DIFT	Ours
Data Probing	68	0.00%	0.00%	**4.60%**
DoS	1,156	55.25%	55.29%	**55.93%**
Malicious Control	178	3.26%	2.14%	**9.61%**
Malicious Operation	161	0.00%	0.00%	0.00%
Normal	69,588	98.91%	**98.92%**	98.90%
Scan	310	0.00%	0.00%	0.00%
Spying	106	0.00%	0.00%	0.00%
Wrong Setup	24	24.24%	**76.00%**	64.29%

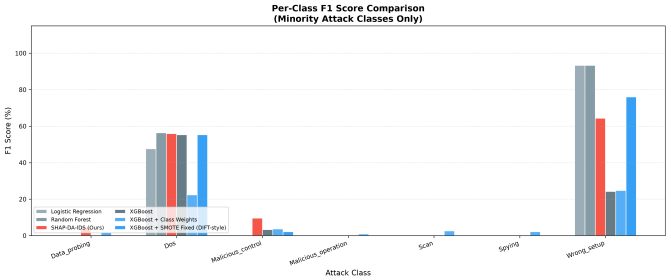


Fig. 5. Per-class F1-score comparison focusing specifically on the seven minority attack categories across evaluated methods.

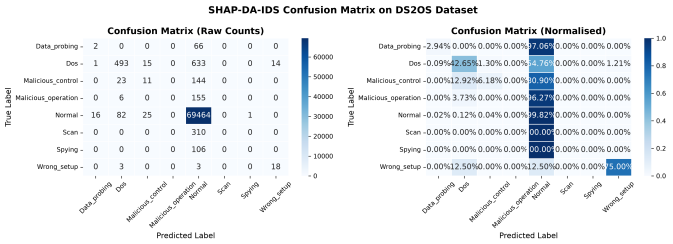


Fig. 4. Confusion matrix of SHAP-DA-IDS on holdout test data: raw counts (left) and normalized classification percentages (right).

C. Analysis of Augmentation Quotas

Table IV and Fig. 7 compare the allocation quotas. Because DIFT clamps minority targets to 1,144, rare attacks remain outnumbered by benign traffic by 243:1. Our adaptive mechanism detects that Data Probing and Malicious Control have preliminary F1 near zero ($D_c > 0.97$) and allocates ~14,500-15,000 samples. In contrast, Wrong Setup (preliminary F1=0.95) receives only 827 samples, avoiding redundant synthesis on easily separable manifolds.

TABLE IV
TARGET ALLOCATION: FIXED (DIFT) VS. ADAPTIVE (OURS)

Class Label	Train N	Diff.(1–F1)	DIFT	Adaptive
Data Probing	274	0.9785	1,144	14,552
DoS	4,624	0.7913	4,624	16,170
Malicious Control	711	0.9734	1,144	14,914
Malicious Operation	644	0.9914	1,144	15,110
Scan	1,237	0.9651	1,237	15,319
Spying	426	0.9729	1,144	14,623
Wrong Setup	98	0.0500	1,144	827

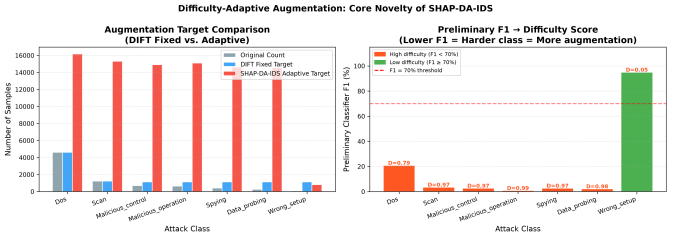


Fig. 7. Difficulty-Adaptive Augmentation mechanism: comparison of sample target allocations (left) and preliminary empirical F1-scores defining class difficulty (right).

D. Threat Attribution via TreeSHAP

Fig. 6 illustrates global TreeSHAP feature attributions on test predictions. The continuous sensor reading (value, mean |SHAP|=1.1502) is the primary predictive driver, capturing anomalous telemetry indicative of data probing and parameter tampering. The invoked command (operation, mean |SHAP|=0.6203) serves as the primary discriminator for active attacks, capturing command loops and unauthorized actuation. Table V lists the top 3 attribution drivers per attack category.

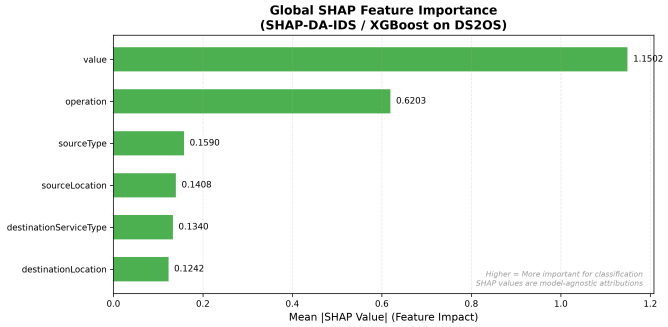


Fig. 6. Global SHAP feature importance computed via TreeSHAP on test predictions, reflecting mean absolute Shapley attribution per feature.

TABLE V
TOP DISCRIMINATIVE ATTRIBUTION DRIVERS PER ATTACK CLASS

Attack Class	Top 3 Discriminative Features (Mean |SHAP|)
Data Probing	value, operation, sourceType
DoS	operation, value, sourceType
Malicious Control	operation, value, destinationLocation
Malicious Operation	operation, value, destinationServiceType
Normal	value, operation, sourceType
Scan	value, operation, sourceType
Spying	value, operation, sourceLocation
Wrong Setup	value, operation, sourceLocation

E. Computational Latency and Deployment Feasibility

Table VI reports execution runtime. SHAP-DA-IDS trains in 11.54s and processes the 71,591 test instances in 167.00ms (2.33µs per flow), confirming readiness for real-time edge IoT gateways.

TABLE VI
COMPUTATIONAL RESOURCE CONSUMPTION ON DS2OS DATASET

Model Configuration	Train Time (s)	Inference Latency (ms)
Logistic Regression	**1.33**	**4.00**
Random Forest	8.63	236.58
XGBoost (Unbalanced)	8.96	162.53
XGBoost + Class Weights	9.16	161.00
XGBoost + Fixed (DIFT)	9.15	158.64
SHAP-DA-IDS (Ours)	11.54	167.00

VI. LIMITATIONS AND FUTURE WORK

Several limitations should be noted. First, because the original DIFT source code was unavailable and full diffusion training requires enterprise GPUs (NVIDIA A100), we utilized an empirical SMOTE oversampler as the generative backend to benchmark difficulty-adaptive allocation. Integrating a lightweight conditional diffusion model on tabular data is an important next step. Second, certain micro-minority attacks (Malicious Operation, Scan, Spying) remained difficult to separate from normal traffic due to significant overlap in tabular features. Third, our analysis operates on flow-level records rather than raw packet payloads, which precludes detecting encrypted or sequential multi-packet evasion tactics.

VII. CONCLUSION

This paper addressed class imbalance in IoT intrusion detection by examining the limitations of the DIFT framework [9]. We introduced SHAP-DA-IDS, combining SHAP-guided feature selection, difficulty-adaptive synthetic augmentation, and post-hoc TreeSHAP threat attribution. On the DS2OS smart home dataset (357,952 records), our approach elevated Macro F1 from 22.71% to 29.17%, quadrupled Malicious Control F1 (9.61% vs. 2.14%), and established detection on previously undetectable Data Probing attacks, while preserving 98.90% F1 on normal traffic with sub-millisecond inference latency (167ms for 71,591 records). Furthermore, our explainability analysis identified sensor payload values and operation command types as primary indicators of compromise, providing actionable security intelligence for operational IoT environments.

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